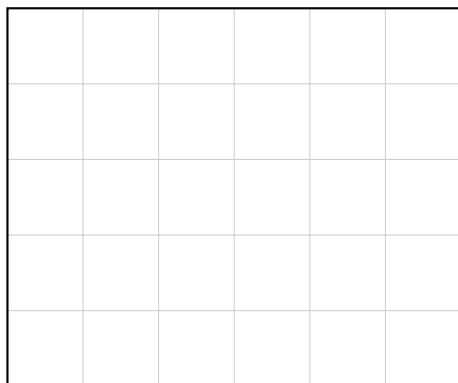
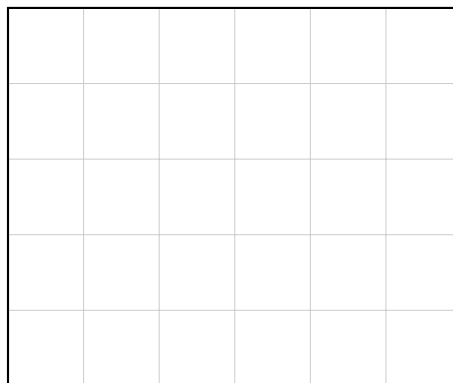


Squares must fill the rectangle, with no gaps, overlaps, or overhang. All edges stay on grid lines. The biggest-square rule removes the largest square from one end of the remaining rectangle, then repeats.

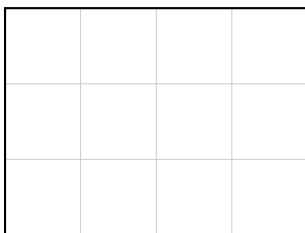
Problem 1: For each rectangle, use the biggest-square rule on the left board. Tile the right board with as few squares as you can, using any whole-number sizes.



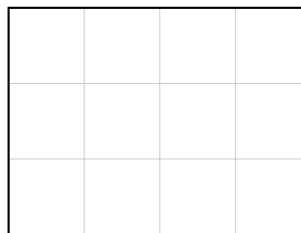
biggest-square rule



fewest squares



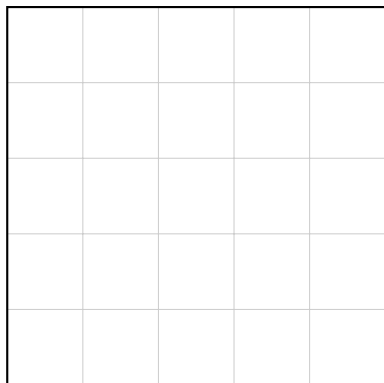
biggest-square rule



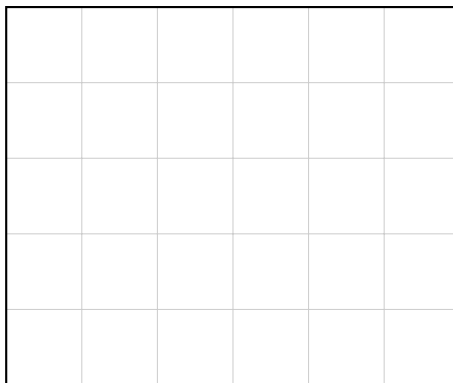
fewest squares

Problem 2: What is the fewest number of squares for the 6×5 rectangle? Explain why a tiling with fewer squares cannot fit.

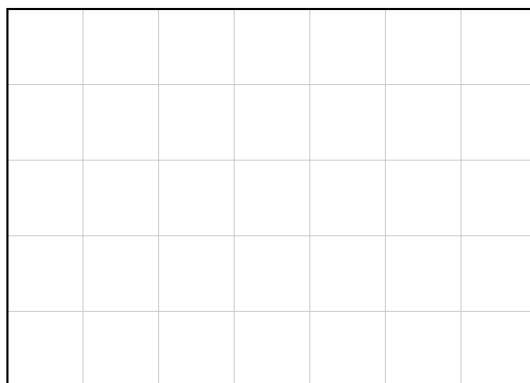
Problem 3: Use only 2×2 and 3×3 squares. Which rectangles can you fill? Draw a tiling for each possible one. Explain why the others cannot be filled.



5×5

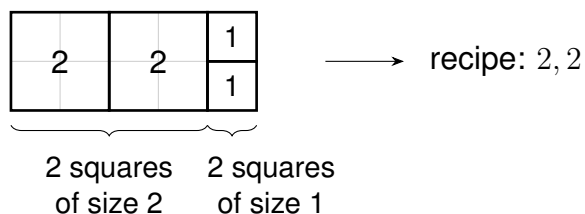


6×5

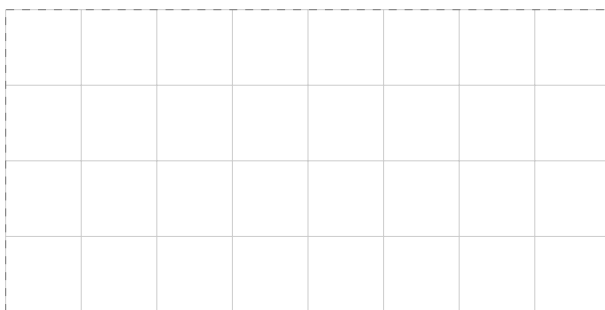


7×5

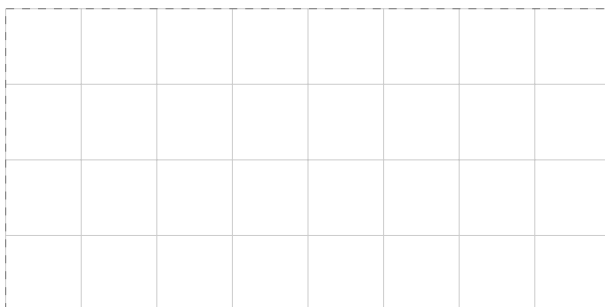
A recipe lists how many consecutive squares of each size the biggest-square rule removes, from largest size to smallest. The sizes are left out.



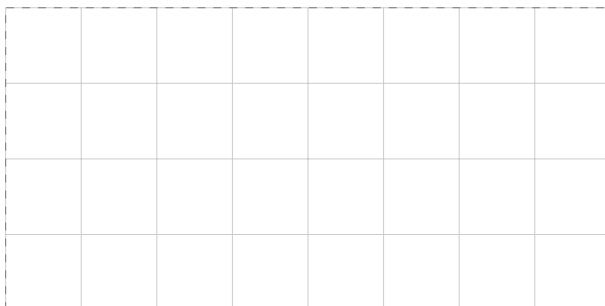
Problem 4: For each recipe, draw a rectangle and its square tiling on the grid. Choose its square sizes yourself, then check it with the biggest-square rule. Can different-sized rectangles have the same recipe? You may use separate graph paper.



recipe: 1, 2



recipe: 2, 1, 2



recipe: 1, 1, 3